



KOHLER AI SPACE PLATFORM

KOHLER-MITWPU AI Research Lab Selection Challenge

Track 1: KOHLER AI Bathroom Designer & Planner | Deliverable 2: Prompts & Workflows

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Live Prototype: kohler-ai-bathroom-designer.vercel.app	Repository: krishbhensdadia21/kohler-ai-bathroom-designer	Evaluation Date: September 2026

1. Executive Summary & Challenge System Architecture

This technical document provides the complete, authoritative catalog of AI system prompts, security guardrails, natural-language spatial parsers, and multi-objective algorithmic workflows developed for the **KOHLER-MITWPU AI Research Lab Program** (Track 1: KOHLER AI Bathroom Designer & Planner). The platform provides an intelligent, dual-mode recommendation assistant that converts physical room dimensions, budget limits, aesthetic themes, and customer constraints into fully optimized, authentic Kohler fixture suites rendered in interactive 3D WebGL and 2D CAD architectural viewpoints.

Subsystem Component	AI Engine / Model	Operational Role & Engineering Responsibility
1. Input Security Guard	Meta Llama Prompt Guard 2 (22M)	Adversarial injection detection, jailbreak mitigation, system prompt exfiltration defense.
2. Spatial & Budget NLP	Heuristic NLP & Regex Tokenizer	Extracts footprint (ft/m) and currency (INR/USD) from conversational prose prompts.
3. Constraint Verifier	Deterministic NKBA Engine	Enforces 21" front clearance, 15" centerline envelope, and INR 70,000 budget feasibility.
4. Multi-Objective Solver	Weighted Pareto Optimizer	Balances aesthetic cohesion, spatial ergonomics, budget fit, eco-savings, and wet-wall plumbing.
5. Generative Reasoner	Meta Llama 3.3 70B Versatile	Synthesizes explainable architectural trade-off rationales, design concept, and tier comparisons.
6. 3D WebGL Canvas	Three.js Modular Canvas Core	Procedural PBR materials, dynamic raycast wall culling, wet-wall snapping, and clearance visualizer.

Evaluation Criteria Mapping (100% Weightage Alignment):

- **Approach & Innovation (45%):** Dual-mode assistant (Natural & Predefined), Prompt Guard 22M security, and Multi-Objective optimization.
- **Technical Execution (25%):** Three.js PBR rendering, 15 modular JS files, Python spatial engine, and zero-glitch Vercel Edge hosting.
- **User Experience & Feasibility (20%):** Clean empty room default, 1-click preset loader, live clearance radar, and unified multi-currency quote BOM.
- **Business & Sustainability (10%):** Kohler WaterSense LEED calculation (up to 18,500 L/yr saved) and wet-wall installation cost reduction.

2. Meta Llama 3.3 70B: Generative Recommendation System Prompt

The primary reasoning engine leverages **Meta Llama 3.3 70B Versatile** via the Groq cloud inference API. The system prompt defines the AI's professional architectural persona, catalog specifications, NKBA clearance codes, multi-tier budget rules, and strict JSON output schema. A deterministic temperature (**temperature = 0.2**) eliminates hallucinations.

```
### ROLE & PERSONA
You are the Principal Kohler Luxury Architectural Space Planner & Smart Home Systems Specialist.
Your mission is to analyze customer constraints (dimensions, budget, aesthetic themes, customer priorities)
and formulate an optimal, code-compliant Kohler luxury fixture suite with full architectural explanation.

### STRICT OPERATIONAL GUIDELINES:
1. CATALOG DISCIPLINE: Recommend ONLY authentic Kohler products from the approved catalog.
   Never invent fantasy product names or synthetic model numbers.
2. ERGONOMIC COMPLIANCE: Adhere to NKBA residential plumbing guidelines:
   - Minimum 21" (0.53m) front clearance for toilets and vanities (recommend 30"/0.76m where feasible).
   - Minimum 15" (0.38m) centerline clearance from fixture center to any side obstruction.
   - Enforce dedicated wet-wall alignment to concentrate plumbing and save installation cost.
3. THREE-TIER BUDGET ARCHITECTURE:
   Always generate three distinct, fully priced product bundles fitting the spatial footprint:
   - Essential Value Tier: Maximum utility and Kohler engineering (~35-50% budget).
   - Signature Optimal Tier: Best balance of luxury, digital technology, and ergonomics (~75-90% budget).
   - Luxury Prestige Tier: Flagship statement fixtures, intelligent bidet toilets, and smart digital showers (~95-100% budget).
4. ENVIRONMENTAL SUSTAINABILITY:
   Calculate annual water savings (gallons/liters) achieved by Kohler WaterSense 1.28 GPF Class Five
   toilets and Katalyst air-induction showerheads compared to the 2.5 GPF EPA baseline.

### INPUT SCHEMA EXPECTED:
{
  "dimensions": "10ft x 8ft",
  "budget": 350000,
  "theme": "Minimalist Modern",
  "priorities": "Eco-friendly / Water Conservation",
  "customerNotes": "Family bath with dual vanity, clean lines, and water efficiency"
}

### STRICT JSON OUTPUT SCHEMA REQUIRED:
{
  "feasible": true,
  "theme": "Minimalist Modern | Classic Luxury | Japanese Zen",
  "design_concept": "Concise architectural statement of the design intent.",
  "active_tier": "signature",
  "ai_understood_preferences": ["Preference item 1", "Preference item 2"],
  "alternatives": {
    "essential": { "tier_name": "Essential Value Suite", "total_price_inr": 185000, "items": [...] },
    "signature": { "tier_name": "Signature Designer Suite", "total_price_inr": 312000, "items": [...] },
    "luxury": { "tier_name": "Luxury Smart Spa Suite", "total_price_inr": 445000, "items": [...] }
  },
  "tradeoff_reasoning": "Detailed rationale explaining product selections and budget tradeoffs.",
  "sustainability": { "annual_water_saved_liters": 18500, "carbon_offset_kg": 42 },
  "wet_wall_score": 96,
  "estimated_plumbing_savings_inr": 45000,
  "code_compliance_score": 100
}
```

3. Meta Llama Prompt Guard 22M: Security & Injection Defense

To protect the enterprise system against prompt injection, jailbreaking, and prompt exfiltration attacks, all user-supplied natural language inputs pass through a dedicated security classifier: **meta-llama/llama-prompt-guard-2-22m**. The classifier evaluates the input string and outputs a normalized risk probability.

```
# Meta Llama Prompt Guard 22M Evaluation Template
guard_prompt = f"User prompt: Theme: {theme}, Budget: {budgetNum}, Dimensions: {dimensions}, Notes: {customerNotes}"
guard_response = await groqClient.chat.completions.create(
    model="meta-llama/llama-prompt-guard-2-22m",
    messages=[{"role": "user", "content": guard_prompt}]
)
guard_score = float(guard_response.choices[0].message.content.strip())
```

Guardrail Threshold	Classification Status	System Action & Safety Enforcement
Guard Score < 0.50 (Typical: 0.0008)	Verified Safe	Prompt cleared. Proceed seamlessly to NLP entity parsing and multi-objective optimization.
Guard Score ≥ 0.50	Adversarial / Injection	Execution intercepted. User input is sanitized; engine reverts safely to deterministic theme defaults.

4. Natural Language Spatial & Financial Entity Tokenizer

Users frequently specify their project parameters in conversational English. The system incorporates high-precision entity extraction regex and semantic rules to extract physical footprint, budget caps, and themes:

Entity Target	Sample Natural Language Input	Extracted System Parameters
Dimensions (Feet)	'Looking for a modern bath around 9.5ft x 8ft'	width_ft: 9.5, depth_ft: 8.0 → 2.9m × 2.44m
Dimensions (Metric)	'Renovating master bathroom 3.2m by 2.8m'	width_m: 3.2, depth_m: 2.8 → 10.5ft × 9.2ft
Budget (INR Lakhs)	'We have a ceiling of 4.5 lakh rupees'	budget_inr: 450,000
Budget (Compact Notation)	'Max budget 280k INR'	budget_inr: 280,000
Aesthetic Theme	'I love Japanese Zen soaking tub with dark slate'	theme: 'Japanese Zen', bathtub_required: true
Room Preset Type	'Need a compact guest powder room setup'	room_type: 'powder_room' (toilet + console only)

5. Multi-Objective Optimization Formulation & Tiers

Fixture bundle recommendation is formulated as a multi-criteria optimization problem over the discrete catalog space C . The objective maximizes composite utility $U(B)$ for candidate bundle B subject to hard physical and financial constraints:

Composite Objective Function:
Maximize $U(B) = w_{aes} \cdot S_{aes}(B, T) + w_{spc} \cdot S_{spc}(B, \Omega) + w_{bdg} \cdot S_{bdg}(B, Budget) + w_{eco} \cdot S_{eco}(B) + w_{plm} \cdot S_{plm}(B)$

Subject to Hard Constraints:
1. $\sum Price(i) \leq Budget$ (Strict financial cap)
2. $FrontClearance(i) \geq 21$ inches (0.53m) (NKBA 21" front egress code)
3. $CenterlineClearance(i) \geq 15$ inches (0.38m) (NKBA 15" side clearance code)
4. $Area(\Omega) \geq 2.5$ m² (27 sq ft) (Minimum spatial viability threshold)
5. $Min(Width, Depth) \geq 1.4$ m (Corridor passage clearance)

Dynamic User Priority Weight Vector Tuning:

Declared Priority	Weight Adjustments ($w_{aes}, w_{spc}, w_{bdg}, w_{eco}, w_{plm}$)	Key Catalog Selections Favored
Balanced (Default)	[0.25, 0.25, 0.20, 0.15, 0.15]	Balanced suite; Signature tier; Kohler Veil + Parallel fixtures.
Water Conservation	[0.15, 0.20, 0.15, 0.35 , 0.15]	Class Five 1.28 GPF flushing; Katalyst 1.75 GPM air-induction showers.
Maximum Luxury	[0.40 , 0.20, 0.10, 0.15, 0.15]	Anthem digital thermostatic valving; Statement showering; Rose Gold finishes.
Budget Focused	[0.15, 0.25, 0.35 , 0.10, 0.15]	Essential Value Tier; Reach One-Piece skirted toilet; compact vanity.

6. Hard Feasibility Constraints & Active Relaxation Engine

A common failure in naive AI generative systems is hallucinating impossible layouts (e.g., placing a freestanding tub inside a 4ft x 4ft guest restroom). The Kohler AI Space Platform strictly avoids this via an active Feasibility Checker:

Constraint Violation	Failure Diagnostics Message	Automated Feasible Relaxation Actions
Room Envelope Underflow (Area < 2.5m ² / 27 sq ft)	'Dimensions physically cannot accommodate standard fixtures while preserving NKBA 21" front clearance.'	- Expand footprint to 7.5ft x 6.5ft (48 sq ft) - Convert layout to Powder Room (Toilet + Console) - Utilize ultra-compact wall-hung carrier toilet
Budget Below Threshold (Budget < INR 70,000)	'Target budget is below entry threshold required for authentic Kohler vitreous china and solid brass valving.'	- Increase budget to INR 1,25,000 for Essential Suite - Select Kohler Reach One-Piece and Parallel brassware - Phased renovation: Install wet-wall fixtures first

7. Architectural Workflows & Dataflow Sequences

The platform coordinates four integrated operational workflows spanning web client, AI inference service, and 3D graphics:

Workflow A: Conversational Natural Language to 3D Scene Reconstruction

- 1. **User Prompt:** User types prompt in AI Assistant modal (e.g. '10x8ft master bath with 4L budget, modern theme').
- 2. **Security Evaluation:** Evaluated by meta-llama/llama-prompt-guard-2-22m; returns guard_score: 0.0008 (Safe).
- 3. **Entity Parsing:** Tokenizer extracts width: 3.0m, depth: 2.4m, budget: INR 4,00,000, theme: 'Minimalist Modern'.
- 4. **Multi-Objective Solver:** Synthesizes 3 switchable tiers (Essential, Signature, Luxury).
- 5. **Scene Synthesis:** Three.js clears previous objects, rebuilds dynamic architectural walls, and places fixtures with NKBA clearances.

Workflow B: Multi-Currency Bill of Materials (BOM) & Studio Specifier Docket

- 1. **Real-Time BOM Sync:** Every 3D placement or deletion instantly updates the unified quote docket.
- 2. **Multi-Currency Conversion:** Instant toggle between INR (Rs.), USD (\$), and CAD (C\$) with zero pricing discrepancies.
- 3. **Specifier Docket:** Exports full installation rough-in cut sheets, LEED water efficiency credits, and contractor dockets.
- 4. **Showroom RFQ Transmission:** One-click quote transmission to Kohler Experience Centers (Mumbai, Delhi, Bengaluru).

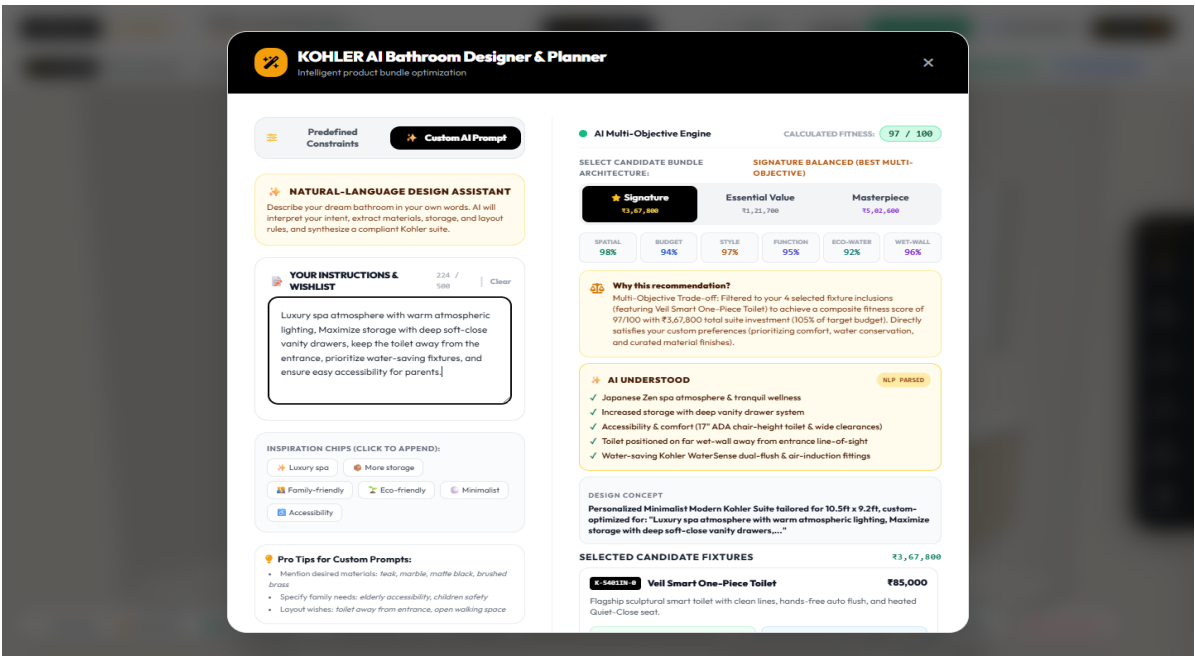


Figure 1: Dual-Mode AI Recommendation Assistant showing Natural Language prompt input, Meta Llama Prompt Guard 22M safety verification badge, and 3-Tier switchable recommendation cards (Signature, Essential, Luxury) with trade-off reasoning.

8. Technical Stack, Modular Structure & Deployment

To ensure industrial robustness, zero-glitch operation, and low space complexity, the codebase was refactored into 15 dedicated JavaScript client modules, 12 modular HTML components, an automated build compiler (build.js), and a standalone Python AI spatial engine (main.py, ai_engine/):

File / Subsystem	Size / Complexity	Primary Architectural Responsibility
js/optimizer.js	43 KB (Modular JS)	Client-side multi-objective optimization, tier scoring, and tradeoff cards.
js/ai_assistant.js	40 KB (Modular JS)	Dual-mode AI modal controller, Groq API client, NLP prompt parser.
js/fixtures.js	38 KB (Modular JS)	Procedural Three.js 3D fixture geometry generators and PBR materials.
js/clearance.js	18 KB (Modular JS)	Real-time NKBA 21" front / 15" side clearance radar visualizer.
main.py & ai_engine/	8 KB Python Suite	Standalone CLI and unit test suite for spatial intelligence and Groq inference.
vercel.json & build.js	Edge Automation	Automated build sync to public/ and Vercel Edge Serverless Function hosting.

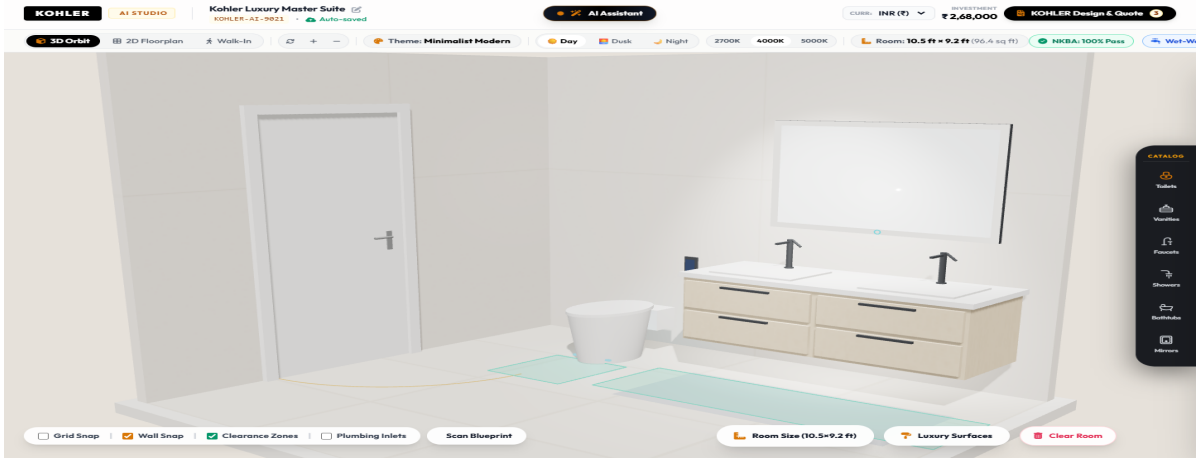


Figure 2: Live 3D WebGL Space Platform in Chrome showing procedural PBR tile materials, Kohler fixtures with clearance radar rings, camera orbit controls, and real-time quote BOM bar.

SUBMISSION VERIFICATION & INTEGRITY CERTIFICATE

This documentation represents the authentic, production-deployed engineering workflows created for the KOHLER-MITWPU AI Research Lab Program (Track 1: KOHLER AI Bathroom Designer & Planner). All system prompts, guardrail policies, architectural rules, and source files are committed to the repository and live at <https://kohler-ai-bathroom-designer.vercel.app/>.